

Computer Science & Information Technology

Database Management System

DPP: 1

Relational model and Normal forms

- Q1** Consider the student relation shown below with schema stud (Sname, Sage, Smail, Smarks),

Stud

Sname	Sage	Smail	Smarks
Rohit	28	R@pw.live	68
Kanika	25	K@pw.live	75
Pankaj	25	K@pw.live	75
Rohit	28	R@pw.live	88
Anjali	26	A@pw.live	75

For the above given instance how many set of attributes of size two can determine each row uniquely?

- Q2** Consider a relation schema $R(A, B, C, D, E, F, H)$ with the given Functional dependency set:

$\{A \rightarrow BC, C \rightarrow AD, DE \rightarrow F, C \rightarrow F\}$

The attribute closure that contains all the attributes of the relation R is?

- (A) AE^+
 (B) CE^+
 (C) AEH^+
 (D) All of the above
- Q3** Consider the following set of FD's:
 $\{V \rightarrow W, W \rightarrow XZ, X \rightarrow YZ\}$ for relation $R(V, W, X, Y, Z)$
 Then the attribute closure of YZ^+ contains how many elements?
 (A) 0 (B) 1
 (C) 2 (D) 3
- Q4** For the given FD set: $\{P \rightarrow QT, Q \rightarrow SU, V \rightarrow U\}$ of a relation $R(P, Q, T, S, U, V)$. Find the set of attributes that is Super key but not a Candidate key?
 (A) PTQ (B) PV
 (C) PQV (D) QV
- Q5** Choose the correct statement from the following.

- (A) The cardinality is defined as the number of attributes in a relation.
 (B) Degree of the relation is the number of tuples in the relation.
 (C) Relation instance is the set of tuples of a relation at a particular instance of time.
 (D) All of the above

- Q6** Choose the correct statements from the following:

- (A) There can be many primary keys for a relation.
 (B) There can be many alternate keys for a relation.
 (C) All the candidate keys are also super keys.
 (D) All the super keys are also the candidate keys.

- Q7** Consider the following statements:

S₁: A key in DBMS is an attribute (or a set of attributes) that helps in uniquely identifying each tuple (or row) in a relation (or table).

S₂: There should be only one candidate key in relation, which is chosen as the primary key.

- (A) Only S_1 is true.
 (B) Only S_2 is true.
 (C) Both S_1 and S_2 are true.
 (D) Neither S_1 nor S_2 is true.

- Q8** Consider the following statements:

S₁: Primary key has no duplicate values it has only unique values.

S₂: Primary key is not necessarily formed using a single column of the table, more than one column of the table can also be used to form a primary key of the table.

- (A) Only S_1 is true.
 (B) Only S_2 is true.
 (C) Both S_1 & S_2 are true.
 (D) Neither S_1 nor S_2 are true.


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Q9 Assume a relation $R(P, Q, R, S, T)$. If PR and RT are the only candidate keys of the relation R , then how many total super keys exist in relation R .

Q10 Assume a relation $R(P, Q, R, S, T, U, V)$. If PQ , RS , and TU are the only three candidate keys of relation R , then how many total super keys exist in relation R ?

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Answer Key

Q1 1
Q2 (C)
Q3 (C)
Q4 (C)
Q5 (C)

Q6 (B, C)
Q7 (A)
Q8 (C)
Q9 12
Q10 74



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Hints & Solutions

Q1 Text Solution:

We can clearly observe that none of the attribute can determine a tuple uniquely (Single attribute), if we check for 2-attribute set then only (Sname, Smarks) can determine a row uniquely for the instance. So, the answer is 1.

Q2 Text Solution:

The attribute closure $AE^+ = \{A, B, C, D, E, F\}$.
The attribute closure $CE^+ = \{C, E, A, B, D, F\}$.
But the attribute H is missing from the closure.
The attribute closure $AEH^+ = \{A, B, C, D, E, F, H\}$.
Therefore, C is the correct answer.

Q3 Text Solution:

The attribute closure of $YZ^+ = \{Y, Z\}$ no other attribute can be determined by YZ^+ . Therefore only 2 elements that is Y and Z are in the YZ^+ closure.

Q4 Text Solution:

The key for the given FD set.
 $\{P \rightarrow QT, Q \rightarrow SU, V \rightarrow U\}$
 $PV^+ = \{P, Q, T, V, U, S\}$
 $PVQ^+ = \{P, Q, T, V, U, S\}$
 $PTQ^+ = \{P, T, Q\}$
 $QV^+ = \{Q, V, S, U\}$
we have PV^+ as the candidate key and also it is the super key. PVQ^+ is the super key but it is not a Candidate Key (not minimal set)

Q5 Text Solution:

- Cardinality is defined as the number of tuples in a relation.
- Degree is defined as the number of attributes in a relation.
- Relation instance is the set of tuples of a relation at a particular instance of time.

Q6 Text Solution:

I. There exists exactly at most one primary key for any relational table while there can be multiple alternate keys for a relation.

II. All the candidate keys are super keys, but it is not compulsory that all super key are candidate keys.

NOTE: A candidate key is minimal set of attributes that determine relational table uniquely. Also, every candidate key is a Super key but every Super key need not be Candidate.

Q7 Text Solution:

S₁: True: A key in DBMS is an attribute (or) a set of attributes that help to uniquely identify a tuple (or row) in a relation (or table).

S₂: False: There can be more than one candidate key in relation out of which one can be chosen as primary key.

Q8 Text Solution:

- Primary key has no duplicate values it has only unique values. Hence S_1 is true.
- Primary key is not necessarily to be a single column more than one column can also be a primary key for the table. Hence S_2 is true.

Q9 Text Solution:

PR	RT	Common
$\downarrow$	$\downarrow$	$\downarrow$
• 2^{5-2}	+ 2^{5-2}	- 2^{5-3}
• 2^3	+ 2^3	- 2^2
• $8 + 8 - 4$	$\Rightarrow 16 - 4 = 12$	

Q10 Text Solution:

PQ	RS	TU	Common between two keys
$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
2^{7-2}	+ 2^{7-2}	+ 2^{7-2}	+ $(2^{7-2} + 2^{7-4} + 2^{7-4})$
Common between all three keys			
$\downarrow$			
+ 2^{7-6}			

